GCI

Introduction to Data Science

松尾・岩澤研究室
MATSUO-IWASAWA LAB UTOKYO

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What is Data Science?

What is Data Science?

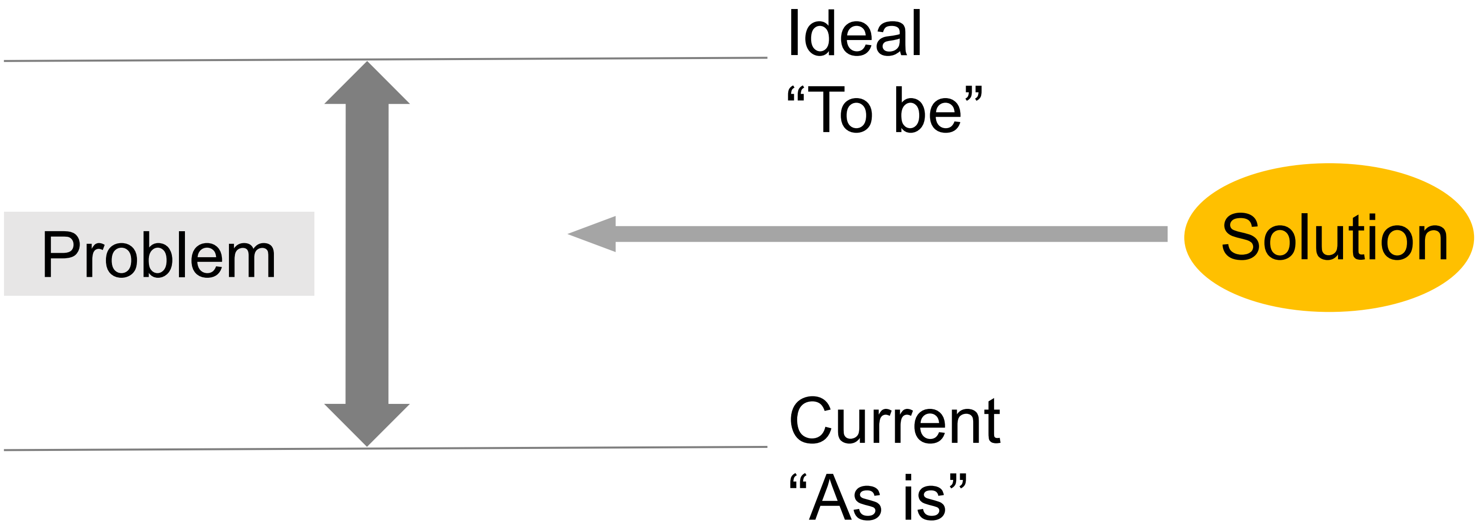
Data Science = Data + Science

Scientific approach* to extract insights from data and solve business problems in the world, converting data into value

(*using mathematics, statistics and IT)

Those who does these jobs are **Data Scientists**

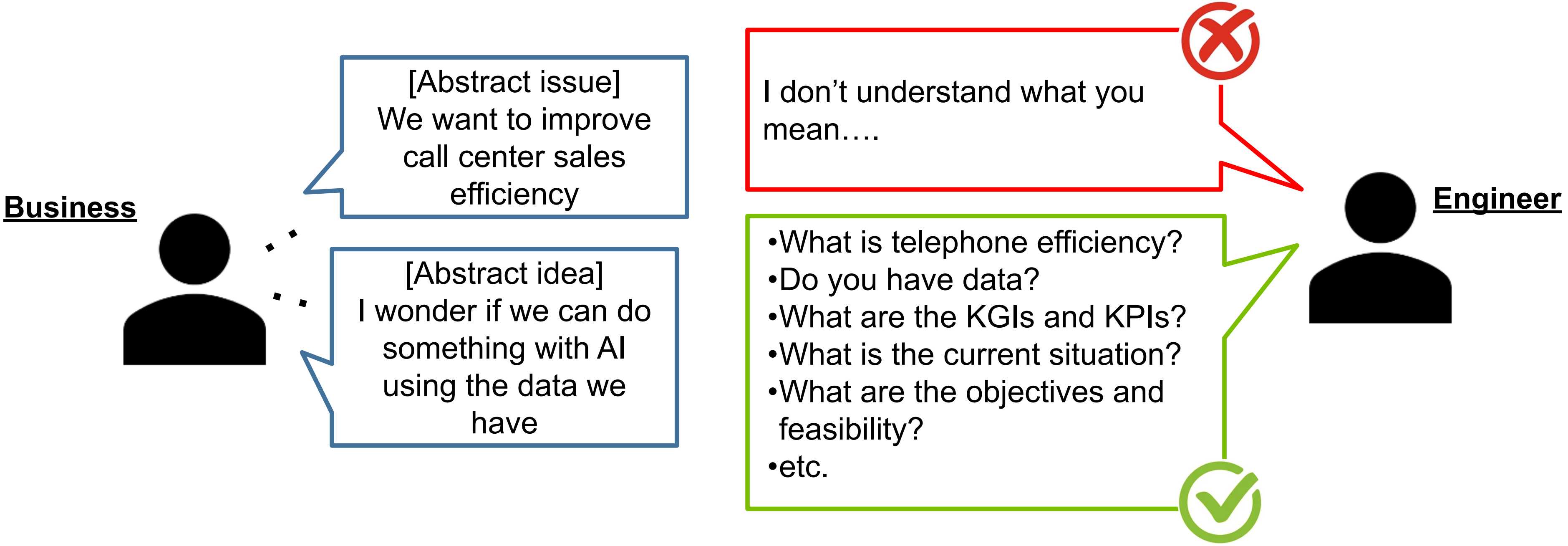
Problem is gap between ideal and current situation



*Reference: “Mondai kaiketsu purofessyonaru 'shikou to gijyutsu’
(Problem Solving Professional ‘Thinking and Technology)’”, Diamond Inc.

It is important to "question" properly

It is important to have ability to define a problem, find a solution and implement it.



Interpret business issues as data analysis and AI issues

- Important to interpret business problems and put them into a data analysis problem
- Tackle large-scale problems by dividing them into smaller problems

[Example]

Improved efficiency of sales calls

Increase sales per call

Increase the order rate per call

Increase order unit price

Increase calls per hour

Reduce time per call

Reduce off-call time

To be: order rate 20%

How to fill up the 5% gap?

As is: order rate 15%

Provide a target list of customers with high contract rates

- Data: attributes, contract status, behavior, contact information, etc.
- Model: supervised learning classification (e.g., logistic regression)

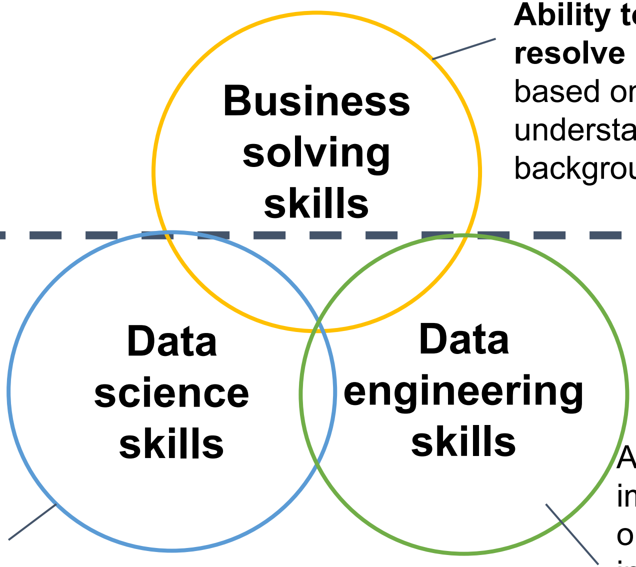
* Reference: “AI sisutemu kouchiku jissen nouhau (Practical know-how for building AI systems”, Nikkei BP

Reference: Job as data scientists

Skill set required of a data scientist

Range of skills to be developed in this course

Practical viewpoints are important too



Ability to organize and resolve business issues based on an understanding of the issue background

Ability to understand and use information science knowledge such as information processing, AI, and statistics

Ability to use, implement, and operate data science in a meaningful way

3 key points in Data Analysis

01	Hypothesis-driven objectives	Business skills, problem finding and solution methods
02	Actionable Insights	Implementation of planning measures, system implementation, etc.
03	Methods Comparison	Statistics skill, data science

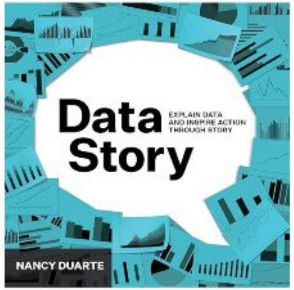
*What are the correct evaluation indicators?
(Are the KPIs and indicators you are using really important?)

Why study Data Science?

- Data Science allows us to discover new perspective and insights
- Learning data science have both personal and professional benefits

Benefits of learning Data Science

- Making informed but unbiased decisions
- “In any role, if you first understand the insights gained from data and then know how to explain them, **your career will advance significantly.**”



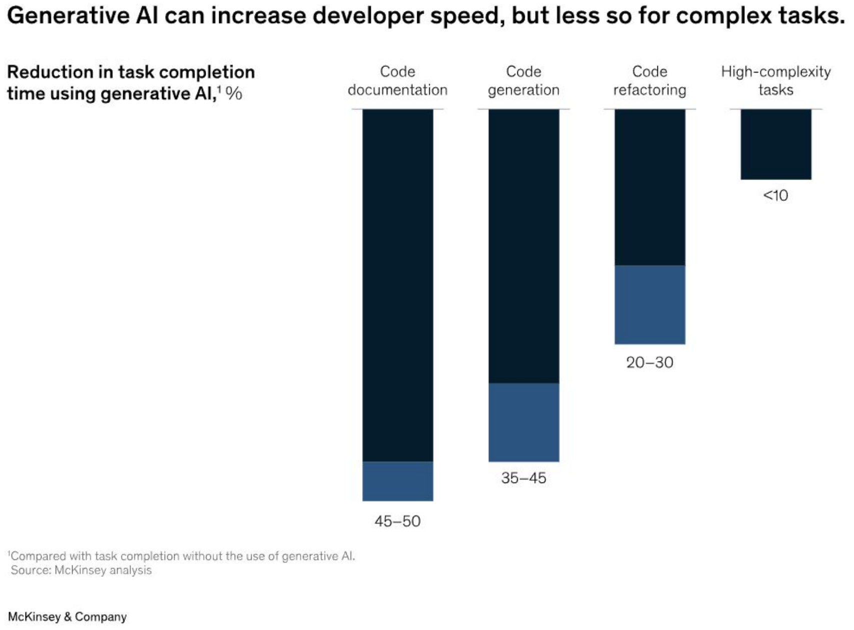
From ”Data Story”, by Nancy Duarte

Using Generative AI in programming

AI cannot fully replace engineer so learning programming is still valuable

Generative AI improve productivity

- Generative AI could increase GDP by 7%, with "software engineering" being one of the major areas with significant cost improvement impacts (ranging from $40 to $50 billion)



https://www.mckinsey.com/featured-insights/mckinsey-explainers/whats-the-future-of-generative-ai-an-early-view-in-15-charts

What is data?

Online

(such as actions on the web)

- Search engines
- Shopping on e-commerce sites
- Social media
- Email
- LINE, Games (such as Pokémon GO)
- Website services (such as web browsing, ad viewing), etc.

Offline

(such as actions in the real world)

- Shopping data from convenience stores and supermarkets
- Public transportation
- Weather data
- Sports (such as baseball, soccer, etc.)
- Medical and healthcare data
- HR data, etc.

Industries that utilize data

Finance
Trading
Risk management
Text analysis of news
Fraud detection

Online Retail Sites
POS data
Web access

Marketing
Web advertising
TV effect
Verification of measures

Games
User behavior analysis

Security
Anomaly detection
Unauthorized access

Manufacturers
Automobiles
Precision machinery

Others
Public, Disaster, Healthcare
Internet, Healthcare,
Broadcasting
Human Resources,
Education, Sports
Law, etc.

Data could be tidy

Affiliation, gender, age, grade, etc.

school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	6	5	6	6
GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	4	5	5	6
GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	10	7	8	10
GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	2	15	14	15
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MS	M	21	R	GT3	T	1	1	other	other	...	5	5	3	3	3	3	3	10	8	7
MS	M	18	R	LE3	T	3	2	services	other	...	4	4	1	3	4	5	0	11	12	10
MS	M	19	U	LE3	T	1	1	other	at_home	...	3	2	3	3	3	5	5	8	9	9

Data also could be disorganized

- Semi-structured financial data, terabyte-level per month, with over 2 billion rows
- 80-90% of data analysis process is spent on data processing before building AI or machine learning models to organize data

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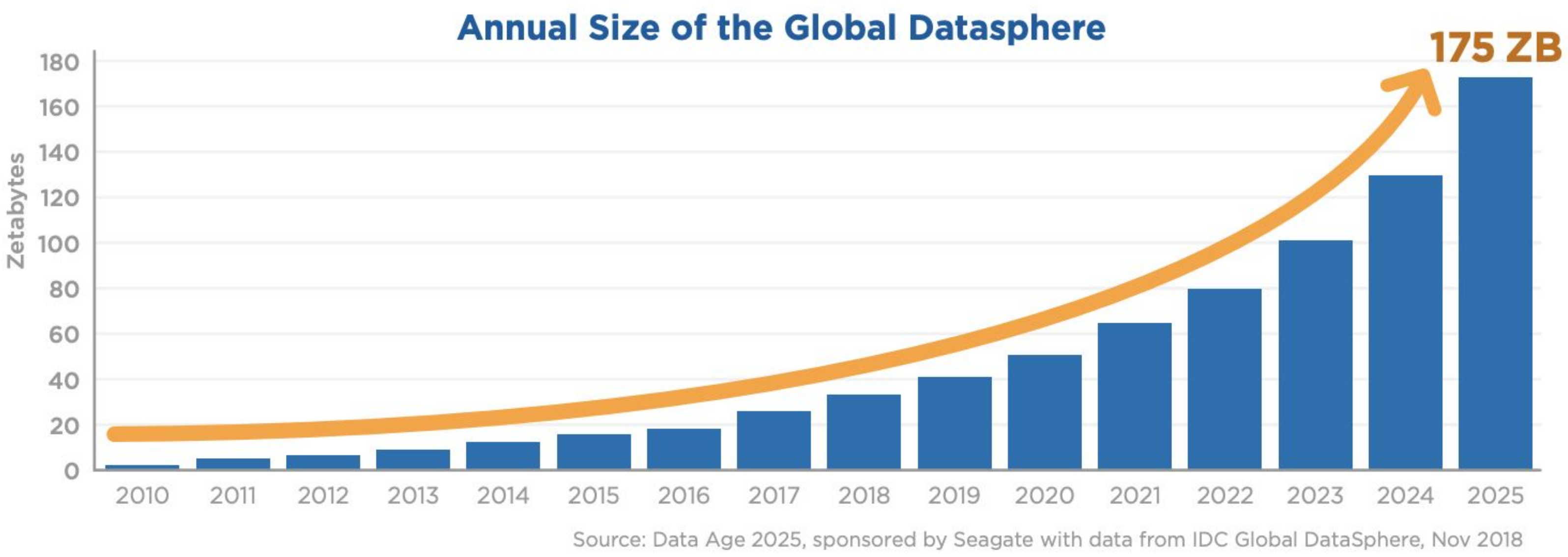
Data is growing exceptionally everyday

Figure 1 – Annual Size of the Global Datasphere



Reference "The Digitization of the World From Edge to Core
(David Reinsel – John Gantz – John Rydning)" 2018 IDC
1 zettabyte is a number with 21 zeros after the 1, equivalent to 1,000,000,000,000,000,000,000 bytes.

Reasons to use data analysis

6 reasons to use data analysis	Keywords
01 Improve decision-making process • Make decision based on data	・Dashborad ・Data and story
02 Understand customer and industry • What type of product are customers using and how do they feel about it etc. • Understanding of industry including competitors	・Increase customer lifetime value (LTV) ・Recommendation
03 Create better product • Smart product • Intelligent home appliances	・Automated driving/delivery ・IoT, home appliances
04 Create better services • Improved efficiency, automated service • Personalized service	・Virtual clinic ・Trend forecast in fashion etc
05 Improve business process • Improved efficiency in business process (marketing, SCM, HR, manufacturing etc)	・Accounting ・Improved efficiency in repetitive tasks
06 Monetize data • Selling data	・Data sales for agricultural machinery manufacturer

Why is data analysis not being done effectively?

Struggling to manage large amount of data in complex system

It is not clear what can be done with the data that is currently available

The data is available, but it has not been organized

We have data but no engineer…

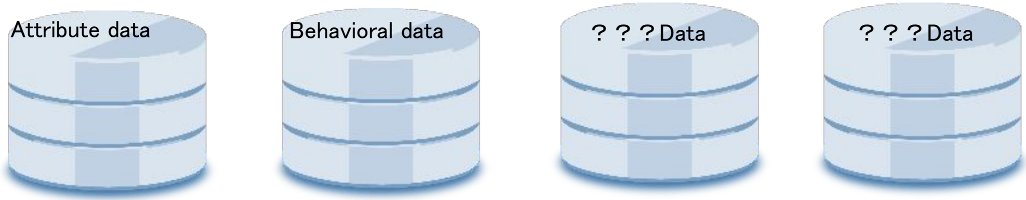
The data simply does not exist!

The purpose of data analysis is unclear

Reference:
”Data analysis failure case collection”, Data bunseki shippai zirei-syu (in Japanese)
Published by KYORITSU SHUPPAN CO., LTD.

It is important to use data strategically

In order to make use of data, it is essential to integrate, process, and manage various customer data (such as attribute data and behavioral data), which is why "data strategy" becomes important.



Data should be collected with a strategy

Data can be both valuable as oil and worthless as garbage

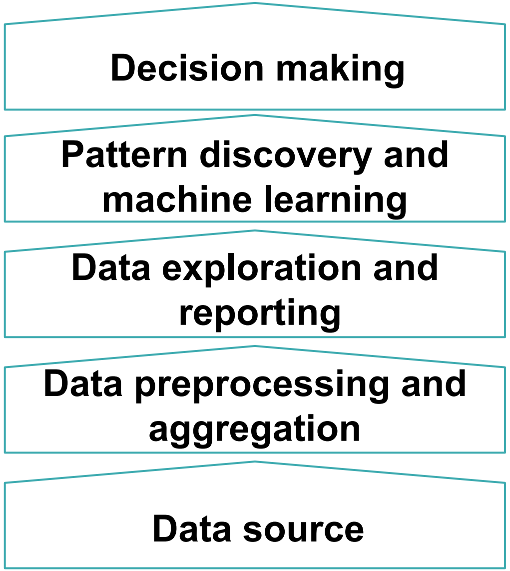
Challenge

The management of old systems, their integration, additional development costs, and the expenses for integrating new systems tend to become substantial.

Data analysis in decision-making

It is important to accumulate data and elevate it to wisdom to be used for decision-making

The potential to support strategic decision-making increases.



Reference: “Data Science” Newton Press

Purpose and strategy in using data is important

It is important to clarify purpose of using the data

Diagram: New data that will be useful for process improvement

Customer satisfaction	• Customer satisfaction level • NPS (net Promoter Score) • Number of complaints, etc	• Information on social media • Sentiment analysis of text data • Customer emotions, etc
Customer Orientation	• Customer attributes • Purchase data • Customer behavior, etc	• NFC data (e.g. transportation IC data) • Weather data • Web access logs • Camera data
Speed	• On-time delivery • Waiting time • Time taken per process, etc	• Machine/sensor/IoT logs • Communication logs • M2M (Machine-to-Machine) data • GNSS (e.g. GPS) data (movement, speed) • Natural/weather data
Quantity & Efficiency	• Productivity per unit time • Inventory turnover rate • Operational rate, etc	
Quality	• Number of defects • Number of errors • Predictive failure rates, etc.	

Reference: Phase in AI and data analysis

Status of data utilization, problems, and measures to be taken

Stage 5: A company with strong analytical capabilities

- Current Status & Challenges: Company-wide data analysis leads to results and keeps the company at the forefront of the industry.
- Countermeasure: Become a leader in analysis.

Stage 4: Companies with limited analytical skills

- Current Status & Issues: Company-wide analytical capabilities are available, but creating new business and differentiation from competitors are issues.
- Countermeasure: Utilize analytical capabilities company-wide in an integrated manner.

Stage 3: Companies strengthening their analytical capabilities

- Current Status & Issues: Started comprehensive data collection and analysis, but still need to understand current trends.
- Countermeasure: Strengthen areas of expertise through data analysis.

Stage 2: Companies with limited use of analytical capabilities

- Current Status & Issues: Data is only partially utilized, so the issue is to improve the analytical capability and utilize it for business improvement.
- Countermeasure: Increase the number of business processes that utilize data analysis.

Stage 1: Companies that lack analytical capabilities

- Current Status & Challenges: Zero use of data is blind, and understanding what has happened in the past is a challenge.
- Countermeasures: Accurately collect data and understand the current situation.

Reference: “Bunsekiryoky wo buki ni suru kigyo (Companies that use analytical skills as weapon”, Thomas H.Davenport, et al.

Hidden data

Decisions based solely on the data we possess are likely to be incorrect

Visible data (database, file, etc)

Hidden data (abstract concepts including those that are not available to make data out of)

Think abstractly about what is not databased, or design something concrete, or go directly to get the information yourself.

Reference:
“Dark Data”, David J. Hand
“Mienai mono wo miru (See the unseen)”, Isao Hosoya

Reference: Failure due to data analysis

Challenge

Whether to change prices to stem the loss of market share from the top

Suggestion

Conducted various analyses and propose keeping price the same

Result

Further decline in market share ranking (from #1)

Overview of GCI

Goal of GCI

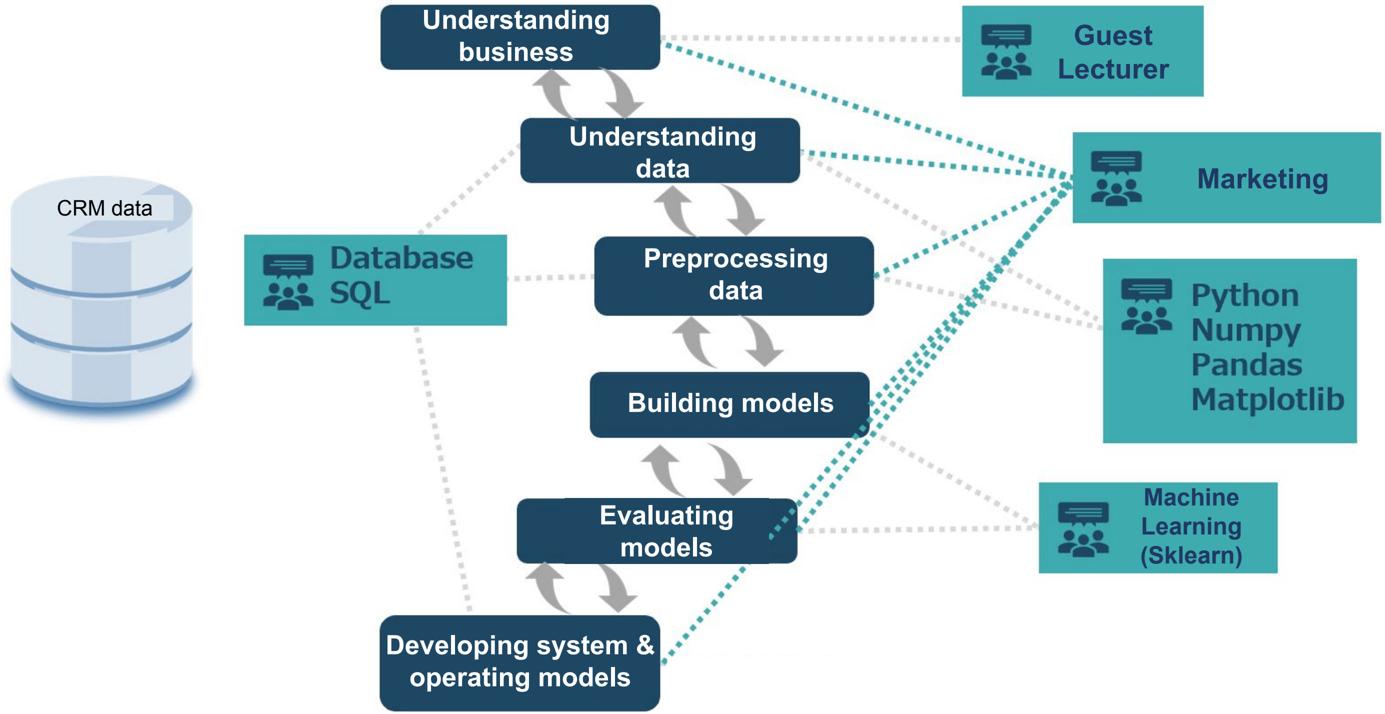
To acquire the basic technical skills necessary for utilizing data in business

01 Implementation skills required for data science (Python, SQL, Machine Learning)

02 Fundamental skills required for data science (DB, etc.)

03 Marketing mindset that is necessary for business (Marketing, external lecturers)

Connection of GCI to real-life data science project



Flow of lecture

Flow

Outline and theory → Explanation of Implementation → Exercise → Explanation

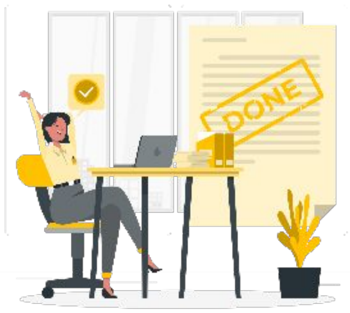
01. Explanation on Slide and Jupyter Notebook (Chapter~.jpynb)

02. Explanation of theory and implementation (code)

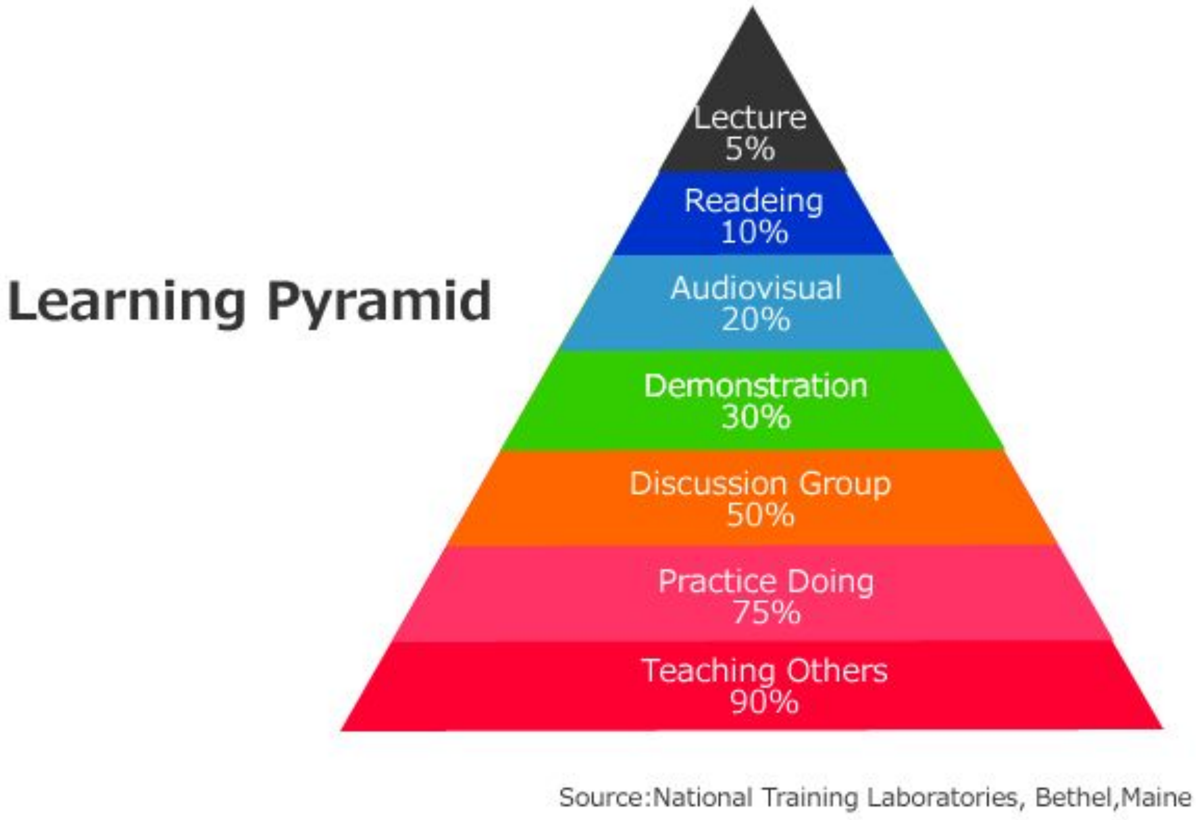
03. Exercise (picked from Jupyter Notebook)

04. Explanation of exercises

(05. Homework at the end)



Learn by outputting



Don’t try to remember everything

Utilize shortcut

It’s like swinging practice

Ask questions if you still don’t understand after 30 mins of research

Research skill

Using google, ChatGPT etc

Learn with your own speed

You will also improve your explanatory skills and connect with other participants

Teach each other

Teaching can also help us understand what we lack understanding on

Prerequisites

- Basic usage of Python, Jupyter NoteBook, Linux
- Ideal if you know:
  - Elementary calculus and linear algebra
  - Basics of probability statistics
- Setting up your own PC
  - We will use Google Colaboratory in GCI so you do not need to install Python
- Make sure to **do some self-study**

Reference: After completing this course

Students	Try working in the field as early as possible (apply to companies that are looking for joint research or part-time jobs)
Working Professional	Find problems in your work that you can Quick Win. + Increase the number of people who can handle technology in your company